Sakuna Harinda Jayasundara

Software & Machine Learning Engineer

Auckland, New Zealand | +64 27 422 5279 | sakunaj1996@gmail.com

linkedin.com/in/sakuna-harinda | github.com/sakunaharinda | sakunah.com

New Zealand Post-Study Work Visa holder — full working rights, no current or future sponsorship required.

Professional Summary

Software and machine learning engineer with 5+ years of industry and research experience building scalable, production-grade systems across backend development, machine learning, and automation. Brings 8+ years of experience in Python, Java, C/C++, TypeScript, and C#, along with extensive work across agentic AI, agentic security, modern ML and MLOps frameworks, CI/CD pipelines, and test automation. Ph.D. from the University of Auckland, with research spanning deep learning, natural language processing, cybersecurity, and HCI. Focused on applying data-driven and research-informed approaches to solve real-world engineering problems.

Technical Skills

Languages	Python, Java, C/C++, Go, C#/.NET, TypeScript, Dart
Backend & APIs	Spring Boot, FastAPI, REST, SOAP, JSON-RPC, microservices
ML & Deep Learning	PyTorch, TensorFlow, Keras, scikit-learn; model fine-tuning, from-scratch architecture implementation (reimplementing layers from papers), NLP, computer vision
LLM & Agentic	LangChain, LangGraph, CrewAI, AutoGen, Google ADK, RAG, Model Context Protocol (MCP)
Data	SQL (PostgreSQL), MongoDB, Neo4j
Cloud & DevOps/MLOps	AWS, Azure, Docker, Kubernetes, Jenkins, MLflow, Airflow, GitHub Actions, Git
AI Dev Tools	Claude, GitHub Copilot, Cursor (daily workflow)

Experience

Doctoral Researcher *Oct 2022 – Jan 2026*

University of Auckland | Auckland, NZ

- Designed ML models based on transformer architecture for transforming natural-language requirements into structured, validated outputs, combining context retrieval, generation, validation, and human-in-the-loop feedback.
- Fine-tuned and continually pre-trained off-the-shelf language models and built deep learning architectures from scratch in PyTorch, including reimplementing model layers directly from scientific papers where no public implementation existed.
- Designed ML-based mechanisms that support automatic error detection and recovery, enabling safe deployment of AI-generated security policies in high-risk environments.
- Conducted empirical user studies with professional policy implementers, demonstrating improved accuracy, efficiency, trust, and adoption potential compared to existing approaches.
- Produced comprehensive analyses and datasets that address long-standing research gaps in access control policy generation.
- Resulted in three publications (one in a flagship cybersecurity conference and two in Q1 computer science journals) with two under review in flagship computer science and HCI conferences (See Google Scholar for the full list of publications.).

Professional Teaching Fellow *Nov 2022 – Present*

University of Auckland | Auckland, NZ

- Prepare and deliver lectures and tutorials for undergraduate and postgraduate courses, including Data Structures & Algorithms, Software Engineering, Cryptographic Management, and Java Programming for Industry.
- Designed course materials and built autograded programming assignments with interactive learning tools, supporting consistent assessment across large student cohorts.
- Supervise examinations and assess assignments and projects, ensuring fair and consistent evaluation.

Software Engineer *Jun 2022 – Oct 2022*

H2O.ai | Remote

- Designed and implemented end-to-end test automation pipelines using Python, Jenkins, and Groovy, reducing manual testing effort and increasing release velocity.

- Led the architecture of a reusable test automation framework, defined the coding standards, and shaped the automation roadmap for cross-team use.
- Integrated automation into CI/CD workflows, improving code quality and shortening time-to-deployment; built and maintained web applications supporting internal and external ML workflows.
- Collaborated with engineering teams to integrate automation into CI/CD workflows, improving code quality and shortening time-to-deployment for key features.

Software Engineer *Jul 2021 – Jun 2022*
Axiata Digital Labs | Colombo, Sri Lanka

- Designed, developed, and maintained backend RESTful and SOAP APIs in Java and Spring Boot for customer-facing services tied to Celcom Malaysia, ensuring high reliability and clean integration with existing telecommunications systems.
- Built an anomaly detection system using OpenTelemetry data and ML models to identify performance and reliability issues in production, enabling proactive detection and improved observability.
- Developed an NLP-based repository classifier to automatically surface relevant public GitHub projects, reducing manual competitive-analysis workload.

Intern Electronics Engineer *May 2019 – Dec 2019*
Paraqum Technologies | Colombo, Sri Lanka

- Developed a GTP (GPRS Tunnelling Protocol) packet analysis tool and test environment for inspecting and debugging mobile core network traffic.
- Built a load-balancing module to monitor network interfaces and manage traffic; optimized existing C++ modules for responsiveness and reliability.

Recent Projects

Ragatouille — Open-Source RAG Guide *2024*
Comprehensive open-source guide to building Retrieval-Augmented Generation systems with LangChain and Neo4j. **Featured on LangChain's official LinkedIn page.** *(Python, LangChain, Neo4j, NLP)*

AgenticSLR — Agentic Systematic Literature Review Platform *2025 – Present*
Web-based agentic platform automating systematic literature reviews to cut researcher workload. *(LangGraph, LangChain, FastAPI, React, TypeScript)*

Custom Model Context Protocol (MCP) Servers *2025*
Custom MCP servers wrapping external APIs with OAuth 2.1 authorisation to extend the capabilities of AI development tools. *(Python, MCP, OAuth 2.1)*

Education

Ph.D. in Computer Science *Oct 2022 – Jan 2026*
University of Auckland | Auckland, NZ

- Researched on human-centric automated access control policy generation using large language models.
- Three publications (one flagship cybersecurity conference, two Q1 journals), two under review. Full list on Google Scholar.

BSc (Hons) in Electronics & Telecommunication Engineering *2016 – 2021*
University of Moratuwa | Colombo, Sri Lanka

- Attained a GPA of 3.92/4.2, obtaining a first class.
- Dean's List Honouree, recognised in all four academic years of the B.Sc. program.
- Developed a deep learning-based maritime surveillance software for the Sri Lanka Navy to detect suspicious activities in the sea using PyTorch and PyQt. The project won the 2nd runner-up in the National ICT Awards for the best university project.

Honours & Awards

- ESET Chillisoft Scholarship in Cybersecurity (2022)
- MBIE Artificial Intelligence for Human-Centric Security Project Scholarship (2022)
- 2nd Runner-Up, National ICT Awards — Best University Project (2021)

References

The references will be provided upon request.